Email Support Agent

Description :

The Email Support Agent is an intelligent system designed to automate the process of responding to customer emails using Generative Artificial Intelligence. In today's digital world, businesses receive a large number of customer emails every day regarding various issues such as order status, payment problems, account access, and general inquiries. Handling these emails manually requires significant time and effort from support teams, which can lead to delays in response and reduced customer satisfaction. To solve this problem, the Email Support Agent system is developed to provide quick, accurate, and professional responses automatically.

Step by step procedure :

Step 1: Start the System

The Email Support Agent application is initialized and ready to accept user input.

Step 2: Receive Customer Email

The system takes the customer email as input through a text box or input field.

Step 3: Read Email Content

The input email is read and stored for further processing.

Step 4: Preprocess the Text

The email text is cleaned by:

- Converting all text into lowercase
- Removing special characters and symbols
- Eliminating unnecessary spaces

This step prepares the data for better understanding.

Step 5: Analyze Email Content

The system examines the email to understand the meaning and purpose of the message.

Step 6: Identify User Intent

The system identifies the type of query such as:

- Password reset issue
- Order status request
- Payment-related problem
- General inquiry

Step 7: Send Input to AI Model

The processed email is sent to the Generative AI model for response generation.

Step 8: Generate AI Response

The AI model creates a professional and meaningful reply based on the email content.

Step 9: Format the Response

The generated reply is structured properly with:

Greeting (e.g., Dear Customer)
Main content (solution)
Closing statement

Step 10: Display the Response

The generated response is shown on the screen for review.

Step 11: Edit Option (Optional)

The user can modify the generated reply if needed.

Step 12: Send Response

The final response is sent to the customer via email.

Step 13: End the System

The process is completed successfully.

Logic / Method Used :

The Email Support Agent works based on a simple input–process–output model.

First, the system takes the customer email as input. The text is cleaned and prepared for analysis. This step removes unnecessary characters and standardizes the format.

Next, the system analyzes the content using Natural Language Processing. The goal is to understand the meaning and intent of the email. Based on keywords

and sentence structure, the system identifies the type of request such as order issue, payment problem, or general inquiry.

After identifying the intent, the processed text is sent to a Generative AI model. The model uses prompt engineering to generate a suitable and professional response.

The response is then formatted properly with greeting, body, and closing statement. Finally, the response is shown to the user and can be sent to the customer.

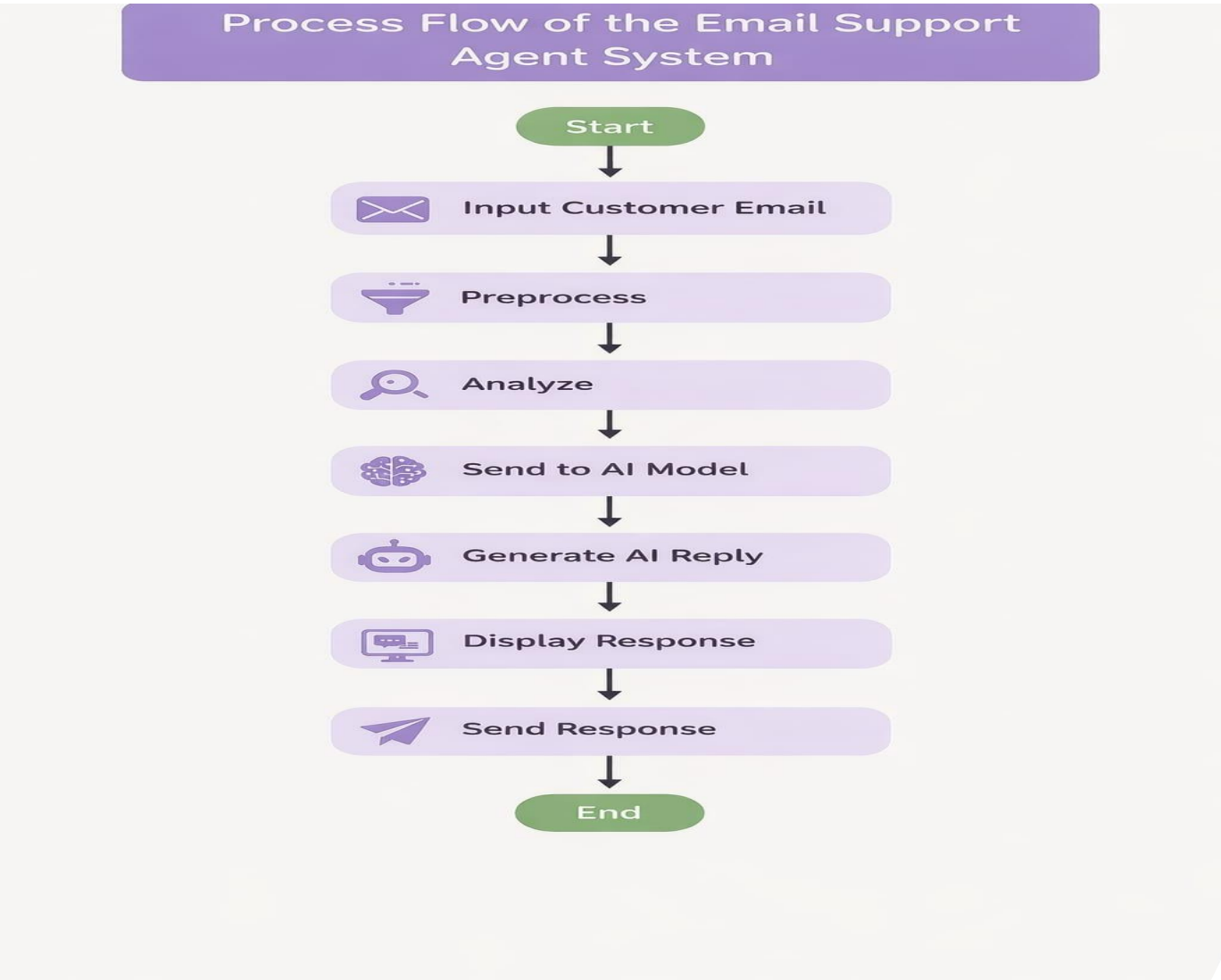
This method ensures automation, accuracy, and faster response time.

Process Flow of the System

Process Flow

The system follows a structured flow from input to output.

First, the user enters the customer email into the system. The system reads the input and processes the text.



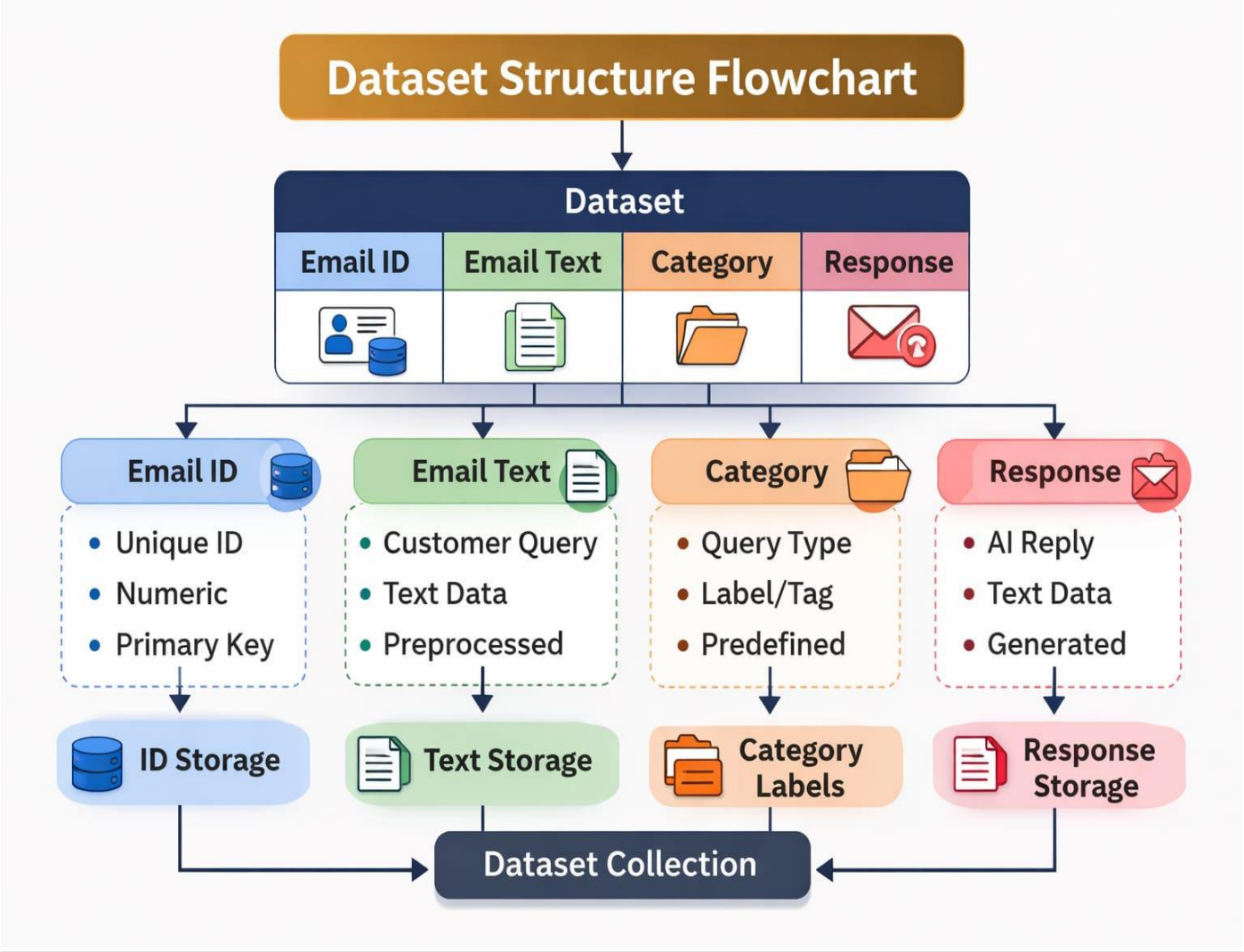
Dataset

Dataset Description

The dataset used in the Email Support Agent project consists of structured email-related data that helps the system understand customer queries and generate appropriate responses. It is designed to train and test the AI model for automatic email reply generation.

Purpose of Dataset

The dataset is used to:

- Train the AI model to understand customer queries
- Classify different types of emails
- Generate accurate and meaningful responses
- Improve automation in customer support systems

Dataset Structure :